

#### LEGEND

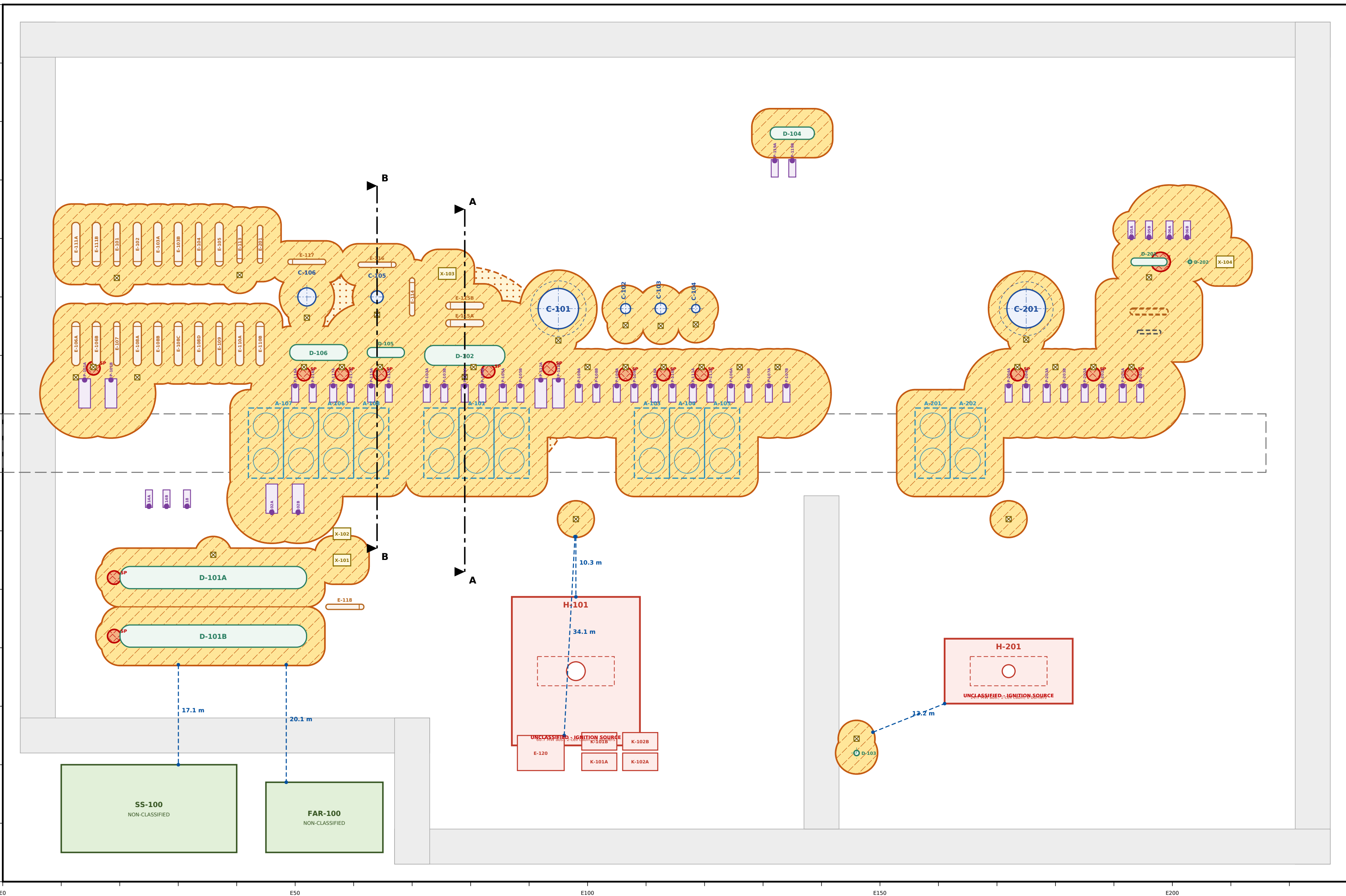
- ZONE 1 - flammable atmosphere likely in normal operation (primary-grade sources)
- ZONE 2 - not likely in normal operation; short duration if it occurs (secondary grade)
- ADDITIONAL ZONE 2, 0.6 m above grade (cat. A / highly volatile liquids, 15 m)
- NON-CLASSIFIED BUILDING (pressurised) / unhatched area = unclassified
- CV station (indicative) SP sample point V atmospheric vent - - - dimension to nearest Zone 2

#### FLUID CATEGORIES

| CAT. | FLUID CATEGORY (EI 15)                                                                                             |
|------|--------------------------------------------------------------------------------------------------------------------|
| A    | flammable liquid which on release vaporises rapidly and substantially (LPG, unstabilised naphtha)                  |
| B    | flammable liquid at a temperature sufficient for boiling on release (crude, naphtha)                               |
| C    | flammable liquid that may be above its flash point or form a mist on release (kerosene, diesel, gas oils, residue) |
| G    | refinery fuel gas / off-gas (may contain H <sub>2</sub> , H <sub>2</sub> S)                                        |
| N    | not a release source                                                                                               |

#### RELEASE SOURCE SUMMARY (full schedule CFU-000-EL-HAC-003)

| SOURCE TYPE                  | CAT | GRP | ZONES             | EXTENT       | EQUIPMENT                                                         |
|------------------------------|-----|-----|-------------------|--------------|-------------------------------------------------------------------|
| Atmospheric vent             | G   | IB  | Zone 1/Zone 2     | 1.5 m; 4.5 m | D-201                                                             |
| Control-valve station        | A   | IA  | Zone 2            | 3 m          | C-105, P-104, P-115                                               |
| Control-valve station        | A   | IB  | Zone 2            | 3 m          | D-102                                                             |
| Control-valve station        | B   | IA  | Zone 2            | 3 m          | C-101, C-106, D-101, P-101, P-116, P-117                          |
| Control-valve station        | C   | IA  | Zone 2            | 3 m          | C-102, C-103, C-104, C-201, E-101, E-106, E-108, E-113, H-101...  |
| Control-valve station        | G   | IB  | Zone 2            | 3 m          | D-103, D-201                                                      |
| Ejector / condenser flanges  | G   | IB  | Zone 2            | 3 m          | J-201, 203 / E-202, 204 (ST-201)                                  |
| Exchanger flanges            | A   | IA  | Zone 2            | 3 m          | E-114                                                             |
| Exchanger flanges            | B   | IA  | Zone 2            | 3 m          | E-116, E-117                                                      |
| Exchanger flanges            | B   | IB  | Zone 2            | 3 m          | E-115A, E-115B                                                    |
| Exchanger flanges            | C   | IA  | Zone 2            | 3 m          | E-101, E-102, E-103A, E-103B, E-104, E-105, E-106A, E-106B, E-... |
| Header box plugs / flanges   | A   | IA  | Zone 2            | 3 m          | A-106                                                             |
| Header box plugs / flanges   | B   | IA  | Zone 2            | 3 m          | A-107, A-108                                                      |
| Header box plugs / flanges   | B   | IB  | Zone 2            | 3 m          | A-101                                                             |
| Header box plugs / flanges   | C   | IA  | Zone 2            | 3 m          | A-103, A-104, A-105, A-201, A-202                                 |
| Pump seal                    | A   | IA  | Zone 2/Zone 2 ext | 15 m; 7.5 m  | P-104, P-115                                                      |
| Pump seal                    | B   | IA  | Zone 2            | 7.5 m        | P-101, P-102, P-103, P-116, P-117                                 |
| Pump seal                    | C   | IA  | Zone 2            | 7.5 m        | P-106, P-107, P-108, P-109, P-110, P-111, P-112, P-201, P-202...  |
| Pump seal                    | C   | IB  | Zone 2            | 3 m          | P-105, P-205                                                      |
| Pump seal                    | C   | IB  | Zone 2            | 7.5 m        | P-206                                                             |
| Sample point                 | A   | IA  | Zone 1/Zone 2     | 1 m; 3 m     | P-115                                                             |
| Sample point                 | A   | IB  | Zone 1/Zone 2     | 1 m; 3 m     | D-102                                                             |
| Sample point                 | B   | IA  | Zone 1/Zone 2     | 1 m; 3 m     | D-101A, D-101B, P-101, P-116, P-117                               |
| Sample point                 | C   | IA  | Zone 1/Zone 2     | 1 m; 3 m     | P-109, P-110, P-111, P-112, P-201, P-202, P-204                   |
| Vessel flanges / instruments | A   | IA  | Zone 2            | 4.05 m       | C-105                                                             |
| Vessel flanges / instruments | A   | IA  | Zone 2/Zone 2 ext | 15 m; 7.5 m  | D-105                                                             |
| Vessel flanges / instruments | A   | IB  | Zone 2/Zone 2 ext | 15 m; 7.5 m  | D-102                                                             |
| Vessel flanges / instruments | B   | IA  | Zone 2            | 3 m          | D-101A, D-101B, D-106                                             |
| Vessel flanges / instruments | B   | IA  | Zone 2            | 4.55 m       | C-106                                                             |
| Vessel flanges / instruments | B   | IA  | Zone 2            | 6.45 m       | C-101                                                             |
| Vessel flanges / instruments | C   | IA  | Zone 2            | 3 m          | X-101, X-103, X-104                                               |
| Vessel flanges / instruments | C   | IA  | Zone 2            | 3.7 m        | C-104                                                             |
| Vessel flanges / instruments | C   | IA  | Zone 2            | 3.85 m       | C-102                                                             |
| Vessel flanges / instruments | C   | IA  | Zone 2            | 3.95 m       | C-103                                                             |
| Vessel flanges / instruments | C   | IA  | Zone 2            | 6.3 m        | C-201                                                             |
| Vessel flanges / instruments | G   | IB  | Zone 2            | 3 m          | D-104, D-201                                                      |
| Vessel flanges / instruments | G   | IB  | Zone 2            | 3.3 m        | D-202                                                             |
| Vessel flanges / instruments | G   | IB  | Zone 2            | 3.45 m       | D-103                                                             |



SCALE 1:500 (A1) - 0 10 20 30 40 50 m

#### CLEARANCE OF IGNITION SOURCES / BUILDINGS TO NEAREST CLASSIFIED AREA

| UNCLASSIFIED ITEM | CLEARANCE m | NEAREST SOURCE | DESCRIPTION                                                 | CHECK |
|-------------------|-------------|----------------|-------------------------------------------------------------|-------|
| H-101             | 10.3        | RS-132         | Control valves FV-1011, FV-1012, FV-1013, FV-1014, FV-1015, | OK    |
| H-201             | 13.2        | RS-126         | Control valves LV-9002 (D-103)                              | OK    |
| E-120             | 34.1        | RS-132         | Control valves FV-1011, FV-1012, FV-1013, FV-1014, FV-1015, | OK    |
| SS-100            | 17.1        | RS-076         | D-101B Electrostatic desalter, 2nd stage                    | OK    |
| FAR-100           | 20.1        | RS-076         | D-101B Electrostatic desalter, 2nd stage                    | OK    |

#### MINIMUM EQUIPMENT PROTECTION (IEC 60079-14 / NEC 505)

| EQUIPMENT               | ZONE 1                                          | ZONE 2                                                                |
|-------------------------|-------------------------------------------------|-----------------------------------------------------------------------|
| LV motors (800 V)       | Ex db or Ex eb (IE, stall relay), IIB T3 (Gb)   | Ex ec (non-sparking), IIA/IB T3 (Gc); AEx ec per NEC 505              |
| MV motors (4.16 kV)     | Not located in Zone 1 (relocate) / Ex pxb       | Ex ec with stator discharge risk assessment / pre-start purge, IIA T3 |
| VFD-fed motors          | Ex db certified with the drive (converter duty) | Ex ec certified for converter duty; T-class verified with VFD         |
| Instruments             | Ex ia / ib (intrinsic safety), Ex db            | Ex ia / ic, Ex db, Ex ec                                              |
| Lighting fittings       | Ex db / eb, IIB T3                              | Ex ec / nL, IIA T3 (IIB in H2S areas)                                 |
| Junction boxes / glands | Ex eb / ec boxes; Ex db barrier glands for Ex d | Ex eb / ec boxes and glands                                           |
| Heat tracing            | Ex 60079-30-1, stabilised design <= T3          | Ex 60079-30-1, self-regulating, T3                                    |

#### NOTES

- Classification per API RP 505 (Zone system) and NFPA 70 Art. 505; release-source extents from RP 505 figures for heavier-than-air vapours in open, adequately ventilated (natural) areas; fluid categories per EI 15 (Model Code of Safe Practice P 15).
- Point-source method (no blanket classification). Areas not shown hatched are unclassified.
- Gas group IIA, temperature class T3 generally; IIB where H<sub>2</sub>S / fuel gas / off-gas is present (overhead system D-102 / A-101 / E-115, hotwell D-201, D-202, D-103, D-104, ejector set, sour-water pumps).
- Fired heaters H-101 / H-201 are ignition sources and are not classified; fuel-gas valve trains are of welded construction with minimum flanges; electrical equipment within 3 m of FG flanges shall be Zone 2 rated. Distances from the nearest Zone 2 boundary are dimensioned.
- PSVs discharge to the closed flare system - no atmospheric release envelope. Drains are closed (to OWS / sour water); below-grade pits, trenches and catch basins within a Zone 2 are Zone 1.
- Pipe rack: only flanges, valves and instrument connections are sources (3 m Zone 2); straight welded runs are not sources. Control-valve stations are shown at indicative locations pending P&IDs / piping layout.
- Pump seals are secondary-grade releases (API 682 seals, open area) -> Zone 2 only; Zone 1 applies to primary-grade sources (sample points, try-cocks, atmospheric vents).
- SS-100 and FAR-100 are non-classified pressurised buildings; HVAC intakes shall be located in unclassified area (>= 3 m from any Zone 2 boundary). Gas detection at intakes.
- Equipment surface temperatures of process plant may exceed fluid AIT (hot oil > 260 C); T-class applies to electrical equipment. Release source schedule: CFU-000-EL-HAC-003; sections: CFU-000-EL-HAC-002.

| REV                                                                         | DATE         | DESCRIPTION              |                    |
|-----------------------------------------------------------------------------|--------------|--------------------------|--------------------|
| A                                                                           | 2026-10-02   | Issued for review (FEED) |                    |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |              |                          |                    |
| HAZARDOUS AREA CLASSIFICATION<br>PLAN - ZONE 1 / ZONE 2 (API RP 505)        |              |                          |                    |
| DISCIPLINE                                                                  | STAGE        | SCALE                    | SHEET              |
| ELECTRICAL                                                                  | FEED         | 1:500                    | 1 OF 1             |
| DRAWN<br>CLAUDE CODE                                                        | CHECKED<br>- | APPROVED<br>-            | DATE<br>2026-10-02 |
| DRAWING NUMBER<br><br>CFU-000-EL-HAC-001                                    |              |                          | REV<br><br>A       |